Muhammad Muntasir

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SUMMARY

Results-driven Backend Developer with 5+ years of expertise building scalable, AI-integrated systems that accelerate delivery and drive efficiency. Specialized in Golang with strong proficiency in TypeScript and JavaScript, I architect high-performance microservices, real-time communication systems (WebSocket, SSE), and intelligent automation workflows. Proven track record integrating AI/ML models into systems from automated scoring and PDF summarization to real-time transcription and smart analytics delivering measurable improvements in speed and user experience. Expert in modern backend frameworks (Gin, Echo, Fiber, NestJS) and event-driven architectures using Kafka and RabbitMQ. I leverage AI tools and automation to optimize development cycles, enabling faster iterations and more robust solutions. Collaborative team player committed to continuous learning and staying ahead of emerging technologies in the rapidly evolving software landscape.

SKILL

Languages:
- **Golang (Primary), TypeScript, JavaScript, SQL**

Frameworks & Runtimes:
- **Gin, Echo, Fiber, Node.js, NestJS, Express, Socket.io**

Architecture & Patterns:
- **Microservices, Event-Driven Architecture (EDA), REST APIs, Clean Architecture, Concurrent Programming, Domain-Driven Design**

Databases & Caches:
- **PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, GORM, Sequelize, Mongoose**

Message Brokers & Streaming:
- **Apache Kafka, RabbitMQ, Redis Pub/Sub**

Infrastructure & DevOps:
- **Docker, CI/CD, Prometheus, Grafana, Loki, Git, Bitbucket, Jira**

Security & Protocols:
- **Keycloak (SSO/RBAC), WebSocket, Server-Sent Events (SSE), WebRTC, End-to-End Encryption**

EDUCATION

STMIK DHARMA NEGARA — Bogor, ID
Bachelor Degree
Major in Informatics Engineering

PROFESSIONAL EXPERIENCE

AJARI Technologies Jakarta
Backend Developer July 2024 – Present

As a Architected and delivered AI-integrated backend infrastructure powering enterprise Learning Management Systems (LMS) and video conferencing platforms for government agencies and corporate clients. Led development of intelligent automation features including ML-powered automated scoring, real-time PDF summarization, AI-driven question generation, and smart anti-cheating detection systems—reducing manual review time by 60% and improving assessment accuracy.

Built production-grade real-time services handling 10,000+ concurrent connections using WebSocket and SSE, including end-to-end encrypted chat systems, push notification engine, and analytics-driven reporting dashboard with complex query optimization. Integrated AI models via RabbitMQ for asynchronous processing and implemented NFC-based authentication for seamless, secure user access.

Developed Elasticsearch-powered search and indexing system enabling sub-second content retrieval across millions of documents. Created intelligent meeting automation system using WebRTC (LiveKit) with AI-powered audio-to-text transcription and auto-generated meeting minutes, accelerating post-meeting workflows by 40%.

Key Projects and Integrations :

- **Learning Management System** for **:**
  - **Direktorat Jenderal Pajak (DJP)**
  - **Kemendagri (BPSDM)**
  - **LearnXpert**
- **Video Conference Engine and Chat Engine** for **:**
  - **SeeU**
  - **Jamdatun**
  - **Workplaze**
- **Recording for Generated Minutes of Meeting** for **:**
  - **Bappenas**
- **End to end realtime chat encryption** for :
  - **Sentinel App**

- Developed and maintained backend systems for AI-powered products including LMS and video conferencing platforms.
- Integrated Elasticsearch for fast and accurate file and content indexing, search, and data retrieval across modules.
- Designed and implemented real-time socket services for secure chat with end-to-end encryption and features such as archive, star, and pin..
- Built middleware for ACL-based permission management across different product modules.
- Developed a real-time Notification Service and Reporting Service with optimized queries for analytics dashboards.
- Implemented NFC-based authentication for secure and seamless user login.
- Created backend support for AI-driven video conferencing, integrating audio-to-text, summarization, and meeting minutes generation.
- Utilized RabbitMQ for asynchronous processing, SSE and WebSocket for real-time updates, and WebRTC via LiveKit for video communication.

Tech Stack: Golang, PostgreSQL, MongoDB, Redis, Elasticsearch, RabbitMQ, WebSocket, Server-Sent Events (SSE), WebRTC (LiveKit), Keycloak, Docker, Prometheus, Grafana, Git, Jira, Bitbucket

STG Auto Group California, US (Remote)
Backend Developer(Project Based) Maret 2024 – July 2024

At STG Auto Group, a used car dealership startup offering good condition. Passed the certified test by a third party to check the car. Provide the user to sell their car, make an offer, negotiation, delivery, and make a deal in app. Provide the result of the inspection report to a customer.

- Trusted to design and build a backend concurrent generate report for generating reports of the leads deals on platform.
- Trusted by the team to design leads deals data using mongoDB database
- Developing in microservice architecture.
- Developing and maintaining robust APIs to support front-end applications and collaborating closely with the team Backend Developer to deliver comprehensive application solutions.

- Developed and Built for the implementation of gmail third party, Using Golang. This involved building a new backend codebase that followed Clean Code Architecture principles.
- Provide the Gmail api in a dashboard of the leads.
- Implement and Build the gmail data by request to make it efficient and more effective in user interface.
- Developed the efficiency of the previous leads and added some features in the dashboard API.
- Actively resolving issues and supporting internal teams when there are data issues on the platform.
- Utilizing an array of technologies, including Golang, MongoDB, phpmyadmin (MySQL), Git for versioning control, Redis, Trello & Github

PT. YesDok Indonesia Jakarta, ID
Backend Developer June 2022 – Maret 2024

At PT YesDok Indonesia, a telemedicine startup offering e-health services across Indonesia, I was assigned as a Backend Developer responsible for designing and developing APIs Golang Microservice to support front-end applications and discussing with the Backend team in designing future features.

- Trusted by the lead to design and build a backend ecosystem for realtime client to client communication using socket.io to create video call based health consultations using client RTC.
- Trusted by the team to design and build real time live monitoring of all consultation calls & video calls using socket.io and stored into reports using mongoDB database
- Developing in microservice architecture.
- Developing and maintaining robust APIs to support front-end applications and collaborating closely with the team Backend Developer to deliver comprehensive application solutions.
- Developed a new architecture Golang microservice for a “Promo” feature in the core application.
- Developed for the implementation of new technologies, specifically Golang, NodeJS and Typescript using frameworks NestJS, for our version 4.0 API. This involved building a new backend codebase that followed Clean Code Architecture principles.
- Led the development and integration with payments of new features for API version 3.0, one notable success being the "Reschedule" feature.Improved the previous refund process and made it possible to reschedule through pending funds via the API.
- Improved the efficiency of the previous code and database execution process in API version 3.0.
- Design and conducting Unit Testing and a practice that wasn't applied in the older version, to improve application stability and code reliability.
- Actively resolving issues and enhancing the performance of the previous API version, whilst also maintaining APIs developed using Lumen (PHP) and Golang.
- Served as the primary point of contact for backend developers to manage API collaboration with hospital partners in Australia..

Tech Stack: Golang, Node.js, NestJS, TypeScript, MongoDB, MySQL, Docker, Redis, Queue Systems, Socket.io, Jest, **Midtrans Payment Gateway**, Git, Jira, Bitbucket

DDTC Indonesia Jakarta, ID
Backend Developer December 2021 – June 2022

At DDTC Indonesia, a taxation company in Indonesia that often wins world-class awards, I work as a Backend Developer who focuses on developing internal server side applications that are needed to make it easier for companies to process taxation and also clients.

- Trusted to develop features and improvise code and sql queries through API for legacy applications such as DDTC News, DDTC Academy, PMS for internal taxation applications.
- Built containers using docker for new microservices to effectively organize project management and maximize time efficiency.
- Active in discussions with the product owner and team at the end of the sprint and also grooming the features that will be made in order to get maximum results.
- Trusted by the head of IT to develop the ddtc news platform server side to complete features and implement clean code architecture.
- Within seven months of my tenure, I successfully built and integrated API endpoints for features that were rigorously tested by a team of qa engineers, contributing significantly to the efficiency of the system.
- Actively resolving issues and enhancing the performance of the previous API version, whilst also maintaining APIs developed using NestJS and Lumen (PHP).
- Utilize various technologies, including Golang, NestJS, Redis, phpmyadmin (MySQL), Git for version control, Docker, Jira & Gitlab.

PT. Sembilan Sembilan Media (KBM App) Bogor, ID
Backend Developer July 2020 – December 2021

At KBM App, a digital literacy company with hundreds of thousands of books, millions of chapters, thousands of authors and 1 million users. I worked as a Backend Developer where I was responsible for developing and re-structuring APIs for new applications using microservice architecture with molecular microservices framework and mongoDB. as well as migrating data from mysql to mongodb.

- Meeting and creating uml diagrams with senior leads for feature development with the new architecture to get the best results.
- Develop and build microservices using molecular microservices framework with the team and leads.
- Empowered by the lead to develop and create user flow in several services such as royalty, books, affiliate, coins, mission, feeds and profile.
- Built containers using docker for new microservices to effectively organize project management and maximize time efficiency.
- Active in discussions with the product owner and team at the end of the sprint and also grooming the features that will be made in order to get maximum results.
- Trusted by the lead to create a flow and development for daily missions where each user will get a reward after doing a mission. In this task I integrate with other service services to get user missions.
- Build queries to process reports on coin user usage history data using the mongodb javascript library mongoose.
- **Architected dual payment gateway integration (Midtrans & Xendit)** for in-app purchases, virtual coin transactions, and premium content access, implementing fallback mechanisms for 99.9% payment success rate
- Developed comprehensive payment orchestration system handling transaction initialization, webhook processing, payment verification, and automated reconciliation for both gateways
- Successfully migrated databases totaling tens to hundreds of thousands of data using molecular microservices from MySQL to MongoDB.

Tech Stack: Moleculer, Node.js, JavaScript, Lumen (PHP), MongoDB, MySQL, Redis, Kafka, Mongoose, Midtrans & Xendit Payment Gateways, Docker, Git, GitHub, Asana

PROJECTS

API POS for Client in UAE Oct 2022

- Developed APIs dedicated to point of sales applications related to orders, merchants and integrated with payment based in UAE.
- Built the database design for the point of sales required as per the client's brief.
- Work for 3 months provided many API that were rigorously tested by a team of front-end engineers, contributing significantly to the efficiency of the system.